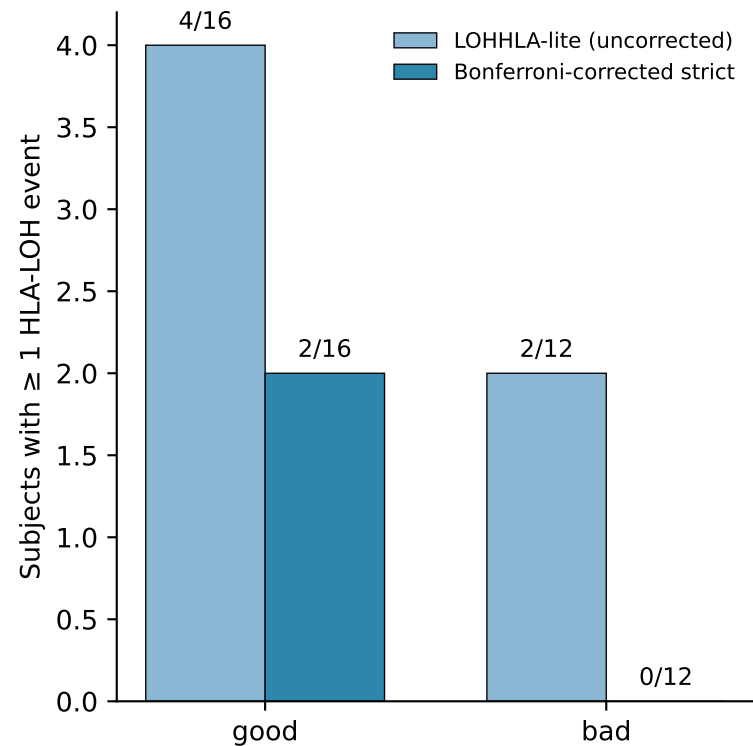
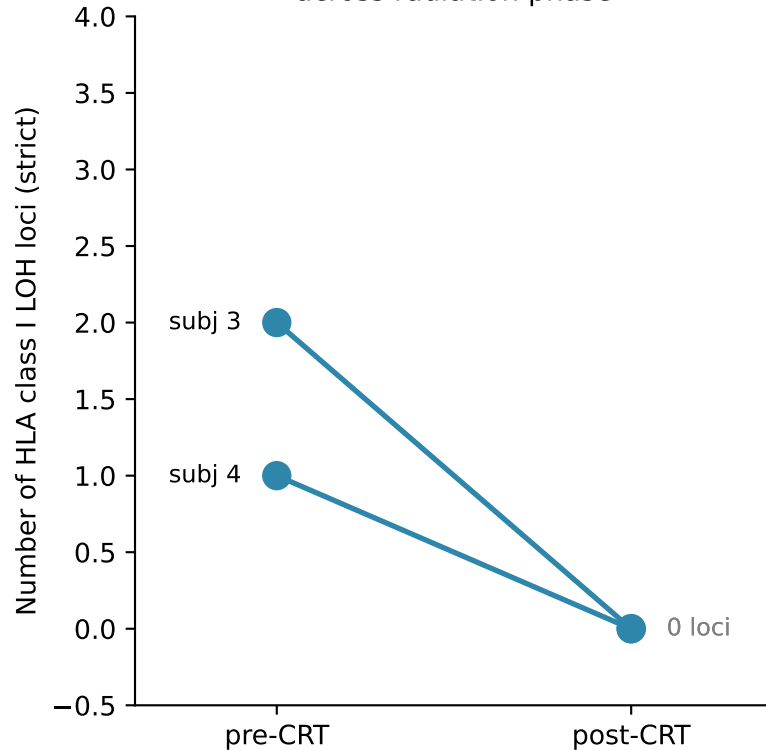


Pre-CRT HLA-LOH prevalence
(lite vs strict criteria)



HLA-LOH clone clearance
across radiation phase



Criteria comparison

LOHHLA-lite (v0.5 draft):

- $|\Delta\text{ratio}| \geq 0.15$
- Fisher $P < 0.05$ uncorrected
- No minimum read-depth filter

Stricter (v0.7 primary call):

- $|\Delta\text{ratio}| \geq 0.20$
- Bonferroni-corrected $P < 0.01$
- Tumor / normal depth ≥ 30 reads each
- Requires normal heterozygosity

Effect of stricter criteria:

- 10 lite events $\rightarrow$ 2 strict events
- Both strict events in eventual good responders (subj 3 and 4)
- Both show complete pre-post resolution of LOH clones

Two-subject nature of strict calls means HLA-LOH clearance is reported as anecdotal corroboration of the cascade model (see Results §3.7), not a between-group statistical claim. Fisher P (good 2/16 vs bad 0/12) = 0.49.